

YULIA VOLKOVA

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I am a MATS 9.1 research fellow working on AI control evaluations. I have 8 years of experience building insider-threat detection and communication-surveillance systems at Behavox for Tier-1 financial institutions. I witnessed firsthand the integration of AI into the regulatory sector, which inspired me to move into AI safety. Further details can be found on my [website](#).

SELECTED PUBLICATIONS

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|------|--|----------------------------------|
| 2026 | Control Evaluations for Covert Malicious Fine-Tuning
Buckworth, T., Moreno, R., Panda, A., Volkova, Y.
Supervised by Dr Francis Rhys Ward. | <i>Submitted to NeurIPS 2026</i> |
| 2026 | Commitment to Cooperation With Self-Negotiated Contracts
Wyse, T., Bustos, K., Volkova, Y. , Kleiman-Weiner, M.
Algoverse AI Safety Research Fellowship, Cooperative AI track. Mentor: Max Kleiman-Weiner. | <i>COLM 2026</i> |
| 2025 | ParaScopes: What Do Language Model Activations Encode About Future Text?
Pochinkov, N., Volkova, Y. , Vasileva, A., Chereddy, S. V. R.
Algoverse AI Safety Research Fellowship. PI: Callum McDougall; mentor: Nicky P. | |

RESEARCH PROGRAMS

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|------------------------------|---|
| Jan 2026 -
September 2026 | MATS 9.0 <ul style="list-style-type: none">Control evaluations for covert malicious fine-tuning under Francis Rhys Ward, with Tyler Tracy and James Lucassen acting as advisorsSelected for the MATS 9.0 Spotlight Talk |
| Nov 2025 | Athena Women in AI Safety Research Fellowship <ul style="list-style-type: none">Mechanistic interpretability approach to monitoring chain-of-thought faithfulness under Arun Jose |
| Oct 2025 | SPAR Fellowship <ul style="list-style-type: none">Chain-of-thought monitorability project under Qiyao Wei, MATS 8.0 and Cambridge University |
| Aug 2025 | Algoverse AI Safety Research Fellowship <ul style="list-style-type: none">Mechanistic interpretability under Nicky P.Cooperative AI under Max Kleiman-Weiner |
| Jul 2025 | ARBOx2 Oxford AI Safety Bootcamp <ul style="list-style-type: none">Completed the ARENA curriculum |

WORK EXPERIENCE

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|-------------------|--|
| Aug 2017–Sep 2025 | Behavox Ltd.
<i>AI-powered compliance and conduct surveillance for financial institutions</i> |
| Jul 2020–Sep 2025 | Senior Data Scientist, ML Development
Key responsibilities <ul style="list-style-type: none">Led the development of Insider Threat ML applications from the Data Science team perspective. These applications analyzed workplace text communications and flagged insider-threat indicators for investigationOversaw and took full responsibility for bringing ML applications into productionDesigned ML applications and maintained end-to-end ownership of their production ML pipelinesManaged technical implementation, including developing Java backend and Groovy components to integrate ML applications into the production systemDirected and oversaw all ongoing production tasks related to ML applications |

Selected technical achievements

- Ran RoBERTa sequence-length experiments that preserved model performance while increasing training throughput by 2× and reducing compute costs by 50%
- Developed a synthetic-data generation pipeline that automatically created and uploaded test datasets to AWS S3 through Jenkins, modernizing the QA process
- Created an automated process that reduced ML application development time from 3 days to 10 minutes
- Consolidated 4 legacy security-monitoring products into 1 scalable solution, reducing backend maintenance requirements

Aug 2017–Jun 2020

ML Applications Developer

Key responsibilities: development of multilingual applications and reports to flag statistical deviations and compliance breaches in client data, designing and building infrastructure to support application creation, QA and maintenance.

- Landed proof of concepts that resulted in 3+ years client contracts
- Improved compliance model recalls from 70% to 90%
- Reduced client bug requests from 1/day to 1/month by setting up an efficient QA process
- Learned Java to remove the Analytics dependency on Backend teams
- Developed best practices for code quality as one of the first Data Science hires and created foundational documentation that helped onboard 13 people

EDUCATION

London School of Economics

MSc Economics

- Distinction in dissertation
- Courses included Advanced Macroeconomics, Econometrics, and Quantitative Methods

London School of Economics

BSc Philosophy & Economics

- First Class Honours
- Courses included Mathematical Methods, Statistics, Econometrics, and Philosophy of Science

EVENTS & TALKS

Aug 2026	GovAI expert panel — “OpenAI x Hugging Face incident: What happened? What’s next?” (Event)
Jul 2026	Secret Loyalties AI safety research hackathon — invited to serve as a judge for the hackathon organized by Apart Research and Formation Research
Jun 2026	MATS project talks at LISA and Google DeepMind London — presented the covert malicious fine-tuning project from Francis Rhys Ward’s stream to research audiences at both offices
Apr 2026	MATS 9.0 Spotlight Talk — opened the London event (YouTube)
Mar 2026	FellowCast Podcast at Constellation — “Covert Malicious Fine-Tuning” (Spotify)

SKILLS SUMMARY

Technology	Groovy, Java, Python, Pandas, Matplotlib, Seaborn, Scikit-learn, NumPy, PyTorch, SQLAlchemy, MySQL, HBase, Kotlin, R, Stata, AWS S3, unit testing, code review, CI/CD, Apache Spark
Data Science	Multivariate classification, SVM, logistic regression, regularization, multicollinearity, feature selection, cross-validation, class balancing, PCA, gradient boosting, t-SNE representations, KNN, and decision trees. Performance metrics: recall, precision, F-score, ROC-AUC, and PR-AUC
Statistics	Hypothesis testing, A/B testing, asymptotic theory, regression analysis, vector autoregressions, time-series analysis, panel-data analysis, fixed and random effects, and IV estimation
Software & Frameworks	Jira, TeamCity, Jenkins, GitLab, Kibana, IntelliJ, PyCharm, Bitbucket, GitHub, JupyterLab, FastAPI, and Airflow
Leadership	Team planning, mentoring, interviewing and hiring, and roadmapping
Other Interests	Oil painting, skiing, and hiking